

# MUHAMMAD ZAIER AHMAD

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## SUMMARY

Software Engineer with a strong systems programming foundation (C/C++, Go) and hands-on full-stack experience across React, Node.js, FastAPI, and Flutter. Currently building a production-grade multi-tenant ERP system with a 400+ table MS SQL Server schema and an orchestrator-driven Node.js backend. Experienced in AI/ML integration, REST API design, and cross-platform mobile development. Seeking to bring both depth and versatility to a high-performance engineering team.

## PROFESSIONAL EXPERIENCE

- Founder & Lead Engineer — Multi-Tenant SaaS ERP (Self-initiated SaaS Product)** *Jan. 2026 - Present*
- Architecting a multi-tenant SaaS ERP with domain-driven schema organization and row-level tenant isolation on a 400+ table MS SQL Server schema covering core business modules, with domain-specific HMS tables planned post-core
  - Engineering a Node.js/Express.js backend using an orchestrator pattern with use-cases, steps, engines, and pipelines for modular, scalable business logic
  - Designing the system as a commercially distributable product beyond single-client deployment, with multi-tenancy and reseller support built into the core architecture

## PROJECTS

- MacroMate [FYP]**
- Developed an AI-powered cross-platform mobile app (Flutter) for calorie, macronutrient, and workout tracking with personalized recommendations
  - Built REST APIs with FastAPI for authentication and data management; integrated Gemini API for food image recognition and a pretrained model for meal plan generation
  - Designed end-to-end from mobile UI to backend, delivering a seamless experience across iOS and Android
- OCR Scanner with AI Summarization**
- Built a Python-based OCR pipeline achieving 95%+ text extraction accuracy on multi-page PDFs with sub-3-second processing time
  - Integrated an AI summarization model to generate document summaries 50% faster than manual reading
  - Engineered using OpenCV for image preprocessing and Gemini API for AI-powered document summarization
- Interpreter in Go**
- Implemented a complete interpreter for the Monkey language in Go, covering lexical analysis, parsing, AST construction, and tree-walking evaluation
  - Applied compiler design principles including Pratt parsing, scope handling, closures, and first-class functions
  - Maintained correctness and reliability throughout using test-driven development with Go's native testing framework

## SKILLS

**Programming Languages:** C/C++, Go, Python, JavaScript, SQL, Dart, Java  
**Backend:** Node.js, Express.js, FastAPI, REST API Design, Orchestrator Architecture  
**Frontend:** React, Redux-toolkit, TailwindCSS, Flutter  
**Databases:** MongoDB, MySQL, Microsoft SQL Server  
**Tools & Frameworks:** Git, Docker, Postman

## EDUCATION

**Bachelor of Software Engineering** *Mar. 2022 - Mar. 2026*  
University of Central Punjab *CGPA - 3.41*

## CERTIFICATIONS

**Supervised Machine Learning** *DeepLearning.AI - Coursera*